



Cream skimming? Evaluating the access to Punjab's public-private partnership programs in education

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ABSTRACT

This paper seeks to evaluate the access to Public Private Partnership (PPP) schools by examining their geographic distribution and the factors that predict whether a child enrolls in a public, PPP, or private school in Punjab, Pakistan using multiple micro data sets. The analysis shows that PPP schools are located in districts where high shares of children are out-of-school, unlike public and private schools. The results indicate that while private school students are the most likely to belong to more affluent households and receive private tuition, there is no evidence that students enrolled in PPP schools come from more advantaged households than public school students. Girls are more likely to be enrolled in PPP schools than in public schools, while students enrolled in the New School Program seem to be the most disadvantaged amongst all PPP students.

1. Introduction

Pakistan continues to struggle with the challenge of low enrollment levels in primary and secondary education. With an adjusted primary net enrollment rate of 77%, there are an estimated 5 million children of primary school age that remain out of school across the country (National Education Management Information System, 2018).¹ In the Punjab-Pakistan's most populous province-the out of school population of primary school age is 1.9 million. In order to address this challenge and the challenge of improving the quality of education in the province, the Government of the Punjab (GoP) implemented a series of reforms beginning with the Punjab Education Sector Reform Programs (I and II), and their successor the 2018 Education Goals. Following the July 2018 elections, the new government has announced its *New Deal* for education in the province, outlining a list of reform priorities for the next five years. One of the major interventions supported through these reform programs and highlighted in the new strategy is scaling up access to education through Public-Private Partnership (PPP) programs. In Punjab, PPPs have been viewed as one of the main mechanisms to bridge the enrollment gap in the province.

Leaving aside whether such PPPs are more or less effective than other types of provision, a more basic question is whether they have improved access and for whom. This paper builds on two strands of research on non-state schools: the first is school location choice and the second is on the access to schooling. In particular, the paper contributes to the literature on PPPs in Education in Punjab, Pakistan by examining

the district level presence of public, PPP, and private schools and the factors associated with attending these different school types by sex. To the best of my knowledge, this is the only paper that examines the geographic distribution of PPP schools and compares and contrasts the factors that predict whether a child attends public, PPP or private school in the province. Unlike previous studies on Punjab, this study will also explore a wider set of variables such as parental education aspirations for their children, and the level of parental involvement in the children's education to identify factors associated with public, PPP and private school attendance. With the drastic scale up of PPPs in Punjab, this research provides necessary evidence to policy makers on the targeting of these interventions. Specifically, this research aims to answer two research questions: *Are PPP programs targeting districts with relatively higher shares of out-of-school children? What are the household and parental characteristics that predict whether a child attends a public school, a PPP school or a private school in Punjab?* The answers to these questions are pertinent because they will provide insight into whether PPP programs are targeting deserving populations in the province. With the recent change in government, this study comes at an opportune time as the School Education Department (SED) is in the process of rethinking its approach to PPPs in Education.

In order to answer the first research question to determine whether PPPs are more likely to be present in districts with high proportions of out of school children, the approach used by Asadullah (2016) is followed by combining administrative data on schools with district level socio-economic information from household surveys. This combined

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¹ The primary school age range used to calculate the net enrollment rate is 5–9 years.

dataset is then used to test whether PPP schools are targeting districts with high proportions of out of school children, while controlling for other district level factors. The study finds that PPP schools appear to have been located in districts where high shares of children are out of school. The second research seeks to identify the factors associated with enrolling in different school types, and is addressed by using a subsample from a school based survey with information on the socio-economic backgrounds of students attending public, PPP and private schools. The study finds no evidence to support the claim that overall, children in PPP schools come from relatively more or less advantaged households than public school students. There is some evidence that two *individual* PPP programs may be enrolling more disadvantaged students than public schools. However, both PPP schools and public schools can do more to target the poorest populations who remain out of school as there is no evidence to suggest that the schools are catering to children belonging to the poorest households in the province. In addition, the study reports that there is no difference in the education aspirations of PPP and public school students. By contrast, private school students are more likely to belong to wealthy households and receive private tuition.

This paper is structured as follows. The next section presents a review of relevant literature on PPPs in education, school location choice, and the determinants of school choice followed by a brief overview of the institutional features of PPPs in the province. Next, the methods used to address the research questions are discussed, along with an introduction to the secondary data used for this study. Lastly, the empirical findings are presented followed by a discussion of the implications of the research along with the identification of areas for further inquiry.

1.1. Literature review

Over the last few decades, the role of the state and the private sector in the provision of education services has become increasingly debated. The modern-day debate initially gained traction when Friedman (1962) advocated for a government-financed, market based education system. Under this proposed system, education vouchers would be issued to families which could be redeemed for educational services. In the United States, this market based approach to education has taken various forms including education voucher programs, charter schools,² and tuition tax credits for households who send their children to private schools (Levin and Belfield, 2003). The underlying rationale of this approach is to break the public sector monopoly in education and create an environment where parents have the ability to choose between schooling options, forcing schools to compete with each other and in doing so, ultimately improve education outcomes. Opponents of the privatization of education argue that such reforms may not address existing inequalities in access to private education, with more advantaged schools being able to recruit higher achieving students, and more affluent households being able to leverage social capital to gain access to better performing schools (Hursh, 2005). As a result, they argue that inequity in access to education would persist and perhaps even exacerbate under such programs.

Despite these criticisms, the push to privatize aspects of education service delivery remains strong with various models being implemented across the world, with mixed outcomes. There is some evidence that education voucher programs improved learning outcomes for voucher students (for example, Angrist et al., 2006; Himmeler, 2007). There is also evidence to the contrary that suggests that after controlling for student sorting or the ability of private schools to select higher achieving students, the positive effects of voucher programs disappear (for example, Hsieh and Urquiola, 2006; Contreras et al., 2008). Aslam

et al. (2017), conducted a review of 22 studies of medium or high quality from developing countries that evaluated education outcomes of Public-Private Partnership programs in education. The authors reported that while there is modest evidence that government subsidies to private and faith based schools can improve learning outcomes, there is some evidence that these programs are reaching poor populations. Similarly, in their review of voucher programs the authors found mixed evidence on the impact of programs on student learning in Chile, India and Pakistan, however they noted the potential for improvements in enrollment particularly for students who would not have otherwise enrolled in school.

Several studies have examined the impact of PPP programs and private schooling on test scores and enrollment outcomes within the context of Pakistan and have found generally positive outcomes (Barrera-Orsorio and Raju, 2015; Barrera-Orsorio et al., 2017; Andrabi et al., 2018; Crawford, 2018). Using an experimental design, Barrera-Orsorio et al. (2017) evaluated the impact of providing subsidies to local entrepreneurs to establish tuition free primary schools in rural areas in Sindh. The authors found that the program increased enrollment by 30% and test scores by 0.67 standard deviations in treated communities. Barrera-Orsorio and Raju (2015) examined the impact of the FAS program on student enrollment and school inputs using a regression discontinuity design by exploiting a key feature of the program design, which requires potential partner schools to participate in a student assessment and meet a minimum score. The authors found large impacts on enrollment, number of teachers and other school inputs for schools near the minimum pass rate as compared to schools that did not qualify for the program but were near the minimum pass rate. A key limitation of this study is that it was restricted to outcomes at the school level, and therefore could not determine whether the program increased enrollment in the communities as opposed to participating schools. Andrabi et al. (2018) tested the impact of providing unconditional grants to private schools on enrollment and student-test scores in 266 villages the district of Faisalabad, Punjab. Using a two-stage stratified randomization design, the authors assigned each village to one of three groups; the *L* arm in which only one private school in the village is given the grant, the *H* arm in which all private schools in the village were provided the grant, and the control group. The motivation for testing outcomes for these two arms was to determine whether the extent of private school market saturation affects equilibrium outcomes of interest. The authors found positive impacts on enrollment for both treatment groups, but only found a positive impact on student learning for the *H* treatment group. Crawford (2018) estimated the impact of contracting out poor performing public schools under the PSSP program on enrollment and test outcomes using a difference-in-difference framework. Crawford found that the program was associated with a large increase in enrollment and a modest decrease in student-test scores. The author could not state with certainty whether the negative impact on student scores was due to a negative treatment effect or a compositional effect due to the change in the student composition in PSSP schools. It is worth noting that the issue of school sorting may persist within this context. In their comparison of learning outcomes of public, private and PPP students in Pakistan, Amjad and MacLeod (2014) found that while PPP students perform better than their public school counterparts, the effect disappears when private tuition is accounted for.

Related to the issue of school sorting and inequitable access to education, is whether non-state schools are located in socio-economically disadvantaged localities. While from a social equity perspective it may be desirable to operate schools in underserved areas, operating in remote localities with limited resources can result in higher transaction costs for operators (Mallick and Nabin, 2018). Therefore, in order to determine the extent to which non-state schools locate in disadvantaged areas, researchers have studied the determinants of non-state school locations and have found mixed evidence depending on the context and type of school (Downes and Greenstein, 1996; Pal, 2010; Asadullah, 2016). Downes and Greenstein (1996) examined private school location

² These are public schools that adhere to their own charter and are often contracted out to education management organizations.

choice in California and found a higher presence of private schools in areas where adults were more educated and that private school locations were sensitive to student population, regional demographic characteristics, and the characteristics of public schools. In a study from India, Pal (2010) found that private schools are more likely to be present in villages that are in close proximity to the district headquarters, where teachers' absence rates are relatively low, and where infrastructural facilities are better. However, Asadullah (2016) reported that non-state schools in the form of Bangladesh Rural Advancement Committee (BRAC), Reaching-Out-of-School (ROCS) project schools and madrassas, have a presence in poor sub-districts. The author found that while BRAC schools did not have a statistically significant presence in areas with a lack of public infrastructure and with low levels of female literacy, ROSC schools targeted more remote, underdeveloped sub-districts and areas that were relatively underserved by government primary schools.

Researchers have also studied the determinants of school choice and the differences in student intake in different school types. In a study from Punjab, Das, et al. (2016) found that the central determinants of school choice include distance to school, school fees, and the characteristics of peers. Alcott and Rose (2015) examined the factors associated with enrollment in specific types of schools in India and Pakistan and reported that in rural Pakistan, private school students were the most likely to belong to wealthy households, to have at least one parent who has attended school, to receive private tuition, and to come from smaller families relative to public school students. Asadullah and Maliki (2018) undertook a similar study of the determinants of enrollment in Islamic, private and public non-religious schools but with additional controls for community characteristics. The authors reported that girls are more likely than boys to enroll in madrassas, boys are more likely to enroll in non-madrassa schools and children from high socio-economic status backgrounds and more developed communities are less likely to enroll in madrassas than other school types.

1.2. PPPs in Punjab

In Punjab, the Punjab Education Foundation (PEF) is a semi-autonomous organization tasked with promoting education service delivery to poor populations through public private partnerships in Education. Currently, there are approximately 2.6 million children studying in PEF partner schools, and to put this number into perspective, there are approximately 12 million students studying in public schools in the province. PEF enrollments have surged drastically between 2010 and 2018, experiencing more than a 100% increase in enrollment during this period. PEF operates three main PPP programs targeted to increase access to education: the Education Voucher Scheme (EVS), the Foundation Assisted Schools (FAS), and the New School Program (NSP). In addition to these programs the Government of Punjab launched the Public School Support Program (PSSP) in 2016, to outsource the management of poor-performing public schools to private sector operators. The EVS was established in 2006 with the objective of offering tuition redemption vouchers to children residing in underprivileged areas in the province to attend low fee private schools. The FAS-PEF's oldest program-offers subsidies on a per student basis to existing private schools in the province, and the NSP offers similar incentives to individuals and organizations to establish new schools in underserved areas. The salient features of these four PPP programs are summarized in Table 1. All of these programs are subject to PEF's quality assurance regime, which consists of quality assurance tests administered to students, as well as regular monitoring of facilities and infrastructure standards. A key concern amongst policy makers regarding these programs is whether they are targeting deserving populations. Underlying these concerns is the notion that PPPs are meant to bridge the enrollment gap in the province, and should target populations that would not otherwise have the means to access schooling.

Table 1
Overview of PPP Programs in Punjab.

Design Feature ^a		EVS	FAS	NSP	PSSP
Objective		To increase access to education to children ages 5-16 residing in underprivileged areas through tuition redemption education vouchers to attend qualifying partner private schools	To promote access to quality education through financial and technical support to partner schools in rural and urban areas. Program provides per student subsidies to already established private schools that meet partnership criteria	To engage private entrepreneurs to establish new schools in under-served localities of the province where there are no government schools within a one kilometer radius.	To outsource education service delivery in poor performing ^b public schools to private operators
Year of Commencement		2006	2005	2008	2016
Monthly Per Student Subsidy Amount		USD 3.5-7 depending on level of schooling	USD 3.5-9.6 depending on level of schooling	USD 3.5-9.6 depending on level of schooling	USD 4.5 for organizations, USD 3.5 for individuals/entrepreneurs
No. of Schools		1,650	3,700	2,404	4,276
No. of Beneficiaries		0.5 million	1.98 million	0.12 million	0.4 million

Notes: (a) Sources: www.pef.edu.pk, Crawford (2018). (b) Poor performing schools include those that are classified as 'over crowded', low enrolment schools, schools with less than 25% of grade 5 students passing exams, and non-functional/closed schools.

2. Methods

2.1. Methods

In order to address the first research question and determine whether PPP schools are more likely to be present in districts with high shares of out of school children, three sources of administrative data were integrated at the district level, along with district level data on the share of primary school age children who were out-of-school (OOSC), and poverty rates which were obtained from the Punjab Multiple Index Cluster Survey (MICS) 2011. A choropleth map was created to compare the number of PPP schools using data from 2017, and the share of the primary school age OOSC population in each district using a lagged measure from 2011. The rationale for utilizing this lagged measure of out-of-school enrollment is to avoid any potential issues relating to reverse causation (i.e. the introduction of PPPs in a district impacting the share of children out of school in that district). Using data from 2011 is appropriate for this purpose because the rapid scale up of PPPs has occurred in the last five years. Following the approach used by Asadullah (2016) to model school count data, Poisson regressions were estimated separately for public, PPP and private schools with district level school counts as the dependent variables while the right hand side variables include the share of children that are out of school, population estimates of the primary school age population,³ the share of the population that lives in rural areas, the share of the population belonging to the bottom two wealth quintiles, and the number of public and private schools in a district. The Poisson regression is appropriate for this purpose since the outcome variable is based on count data and this approach has been utilized in similar studies (Downes and Greenstein, 1996; Asadullah, 2016). The rationale for including counts of public and private schools in the regression is to test whether PPP schools are likely to be present in districts with a relatively limited supply of public schools (after controlling for school age population), and to test whether PPP schools tend to be concentrated in districts where there is a large private sector presence.

To address the second research question and learn more about the characteristics of children that attend public, PPP and private schools, descriptive statistics were calculated to compare the profiles of students by school type. However, relying on descriptive statistics alone to identify factors associated with attending a particular school type is problematic due to the possibility of spurious relationships. For example, when exploring the relationship between the type of school children attend and the number of siblings they have, the true nature of the association may not be apparent until other factors such as parental education and household wealth are controlled for (Alcott and Rose, 2015). To address this, studies have utilized multinomial logit models to estimate the association between independent variables and the probability of enrollment in a particular school type (relative to a baseline outcome such as public school attendance) while holding other variables constant. This approach is particularly appropriate when there is no clear ordering of the dependent variable (which is the case in modeling the type of school attended), as it does not impose a particular structure on the outcome variable as opposed to an ordered logit which assumes a ranking of categories (Long 1997 as cited in Nguyen and Taylor, 2003). In this study, the dependent variable is coded 1 if enrolled in public school, 2 if enrolled in a PPP school, and 3 if enrolled in a private school. A multinomial logit regression thus allows for estimating the impact of a set of explanatory variables on the probability of enrolling in a particular school type relative to public school enrollment (the baseline outcome). Multinomial logit coefficients are estimated as relative probabilities which are not straightforward to interpret, therefore following the approach in studies utilizing similar methods

(see Nguyen and Taylor, 2003, and Asadullah and Maliki, 2018), actual probabilities in the form of average marginal effects (AMEs) are estimated. For this study, AMEs are presented which reflect the change in probability of being enrolled in PPP or private school associated with a one unit increase of an explanatory variable⁴ (Greene, 2010). To calculate AMEs, marginal effects are calculated at all observed values of a particular variable and then averaged across effect estimates (Asadullah and Maliki, 2018).

2.2. Data

The data sources used to examine the geographic distribution of schools are the Annual School Census 2017 of public schools, PEF's administrative data on partner schools from 2017, the Private School Census Data from 2016, and the MICS 2011. Table A1 presents summary statistics generated from this combined data set. The data source to answer the second research question is the Punjab Service Delivery Indicator Survey (PSDIS) commissioned by the World Bank in Punjab, Pakistan. This is a school based longitudinal survey that consists of two rounds of data collection in April/May 2018, and November/December 2018. The first round of data collection covered 812 public, PPP and private schools in the province in both urban and rural localities, while the second round of data collection covered 200 schools (a subset of the original sample surveyed). The survey was administered in six districts in the province, one district from the northern region, three districts from the central region, and two districts from the southern regions of the province.⁵ In order to draw a representative picture of all primary schools in the Punjab province, a sampling frame had to be created using administrative data on Public schools, PPP schools, as well as data from the 2016 Private School Census. This sampling frame consists of a total of 118,691 schools with at least one student studying in primary classes. The sample was stratified based on the three main types of schools (i.e. public, PPP and private schools), as well urban/rural classification. In order to create overlap with a sample from a long running school based longitudinal survey, the Learning and Educational Achievement in Punjab Schools (LEAPS) survey, a separate stratum was created for LEAPS schools. Within the strata, schools were selected proportional to size, using the total primary level enrollment.

This paper uses a subset of data from round one of the PSDIS, for which socio-economic information of students was collected. The total sample of students for this study is 2917 grade four students consisting of 1531 public school students, 954 PPP school students, and 432 private school students.⁶ The variables included in the analysis include student sex, student age, in addition to dummy variables measuring mother and father's education,⁷ father's employment status, and whether the child lives in a southern district. Additional dummy variables measure whether the child receives tutoring, whether the parent speaks with the child's teacher,⁸ how often the parent asks the child about his or her learning,⁹ and whether the child reads books at home (other than textbooks).¹⁰ A dummy variable capturing parents' university aspirations for their children is also included.¹¹ Lastly, an asset index score

⁴ For a dummy variable the AME is the difference in probability of being enrolled in a particular school type between the categories coded 1 and 0.

⁵ Districts are Attock, Faisalabad, Lahore, Sargodha, Muzaffargarh and Rahim Yar Khan.

⁶ Private school students are those that attend for profit private schools. This category does not include students who attend NGO schools, madrassas, 'other private schools' and not for profit schools.

⁷ Father and Mother are coded as 1 if they received any schooling, 0 if not.

⁸ This variable is coded as 1 if the parent speaks to the child's teacher at least several times a month, 0 if not.

⁹ This variable is coded as 1 if the parent asks the child about his or her learning at least several times a month, 0 if not.

¹⁰ This variable is coded as 1, if the parent has observed the child reading books at least several times a month, 0 if not.

³ Estimates derived from 2017 Population Census and Pakistan Social and Living Standards Measurement surveys (2014–15).

was created as a measure of wealth using Principal Component Analysis (PCA).¹² This measure of SES is in line with the methods used in related research on Pakistan (see De Talancé, 2017; Das et al., 2016; and Bau and Das, 2017). It is important to note here that there are other supply and demand side variables that are not available in this dataset, which have been found to be statistically significant predictors of school choice in related research, such as village level characteristics, supply of schools and information on birth order for example (Asadullah and Maliki, 2018). A dummy variable capturing urban/rural location was also excluded from the analysis due to the high numbers of cases with missing information particularly for PPP students. The implications for not including these additional controls are discussed in the limitations section of this paper.

2.3. Sample selection

Given that socio-economic data was not collected from all students surveyed in round one, the key concern is determining whether the subset of students for whom socio-economic data is available is truly representative of the larger sample. To test this concern, basic descriptive statistics for students with socio-economic data are compared with students for whom this information was not collected. T-tests with unequal variance were conducted to determine whether the difference in mean values for these groups across select variables for which information was available, were statistically significant for each school type (i.e. public, PPP and private school students) (see Table A2). The variables on which the t-tests were conducted include the share of students living in urban areas, the share of students living in southern districts, age, sex, the class in which the student joined the school, and the Urdu, Math, English, listening, and overall student-test scores. It is evident that across all school types, there is a rural bias, as well as an over-presentation of students living in southern districts in the sample for which socio-economic information was collected. For PPP students, there was a significant difference in the share of students that are female between the sample of students for whom SES data was collected ($M = 0.514$, $SD = 0.500$) and the sample for which there was no SES data ($M = 0.471$, $SD = 0.499$); $t(1865) = -1.977$, $p = 0.048$. There was also a significant difference in the class in which the child joined the school for the sample with SES ($M = 1.254$, $SD = 1.271$) and the sample without SES information ($M = 1.164$, $SD = 1.218$); $t(1799) = -2.574$, $p = 0.010$.

A key question is whether the sample without SES is more or less disadvantaged than those for whom we have SES. One way to assess this is to focus on test scores since these are highly correlated with SES. While there is also a statistically significant difference in Urdu test scores for PPP students with SES information ($M = 0.445$, $SD = 0.152$) and without SES information ($M = 0.438$, $SD = 0.151$); $t(1851) = -2.184$, $p = 0.029$, this may have been down to the southern bias discussed above. There is no statistically significant difference in Math, English and Listening test scores between these two groups of students which is reassuring in that regard. For public school students there are statistically significant differences in Math, Urdu, and English test scores for students with SES information and no SES information, with students with SES information having on average higher math ($M = 0.356$ $SD = 0.146$), Urdu ($M = 0.391$, $SD = 0.164$) and English Scores ($M = 0.381$, $SD = 0.146$) as compared to students without SES information (math score; $M = 0.342$ $SD = 0.142$), (Urdu score; $M = 0.375$ $SD = 0.160$) (English score; $M = 0.372$, $SD = 0.145$).

Interestingly, for private school students, there is a statistically significant difference in English test scores for students with SES ($M = 0.461$, $SD = 0.161$) and without ($M = 0.482$, $SD = 0.167$) with higher test scores for students without SES information; $t(991) = 2.453$, $p = 0.014$. These statistics suggest that the sample of public school students for which SES information is available may not be a truly representative picture of the larger sample, with over-representation of relatively higher achieving students, and those living in rural areas and in southern districts of the province. For PPP students the situation is slightly more reassuring, even though the sample of students with SES information may be more likely to come from rural areas, live in southern districts, and be female than students without SES information, there is no statistically significant difference in student-test scores except in Urdu. For private school students, while the sample of students with SES information may be more likely to come from rural areas and southern districts, and have lower English test scores than students without SES information, the SES sample does not seem to over-represent higher achieving students, as measured by Math and Urdu.

3. Findings

3.1. Are PPP Programs targeting districts with relatively higher shares of out-of-school children?

Overall, public schools account for approximately half of primary schools in the province, while private schools and PPP schools account for one third and 18%, respectively. When examining the correlation between the number of PPP schools in a district and the lagged share of the primary school age population that is out of school, there does seem to be a strong positive relationship $r_s(34) = 0.680$, $p < 0.001$. This relationship is visually represented in Fig. 1 which presents a district-level map of PPP schools in the province and the percentage of the primary school age population that is out of school. The green shaded rectangles represent the number of PPP schools in a district, while the colored districts represent the proportion of the primary school age population that is out of school.

In order to test empirically whether there is a relationship between the number of PPP schools in a district and the share of population that is out of school, Poisson regressions are estimated and the results are presented in Table B1 by school type. Given the small sample size ($n = 36$), two models are estimated for each school type—a basic model with three covariates, and the expanded model with six covariates. Model 1 (the basic model) includes the share of children that are out of school, the estimated school age population and the share of the population in the district that resides in rural areas. Model 2 (the expanded model) includes the same set of variables in model 1 and a measure of district level poverty¹³ and the number of public and private schools in the district. The results from model 2 reflect that the coefficient on the share of children out of school is only positive and statistically significant for PPP schools suggesting that PPP schools are targeting districts with relatively higher shares of out of school children after controlling for other variables in the model. The coefficient on the number of public schools in the district is positive and statistically significant suggesting that there is a positive correlation between the count of PPP schools in a district and the number of government schools however, the coefficient estimate is very close to zero. This is also the case for the coefficient on the number of private schools in a district which is statistically significant but with a coefficient of less than 0.01. There is also no association between the number of private schools in a district and the share of out of school children in a district.

¹¹ Coded 1 if parent's expect their child (or the child themselves) has aspirations for attending university, 0 if not.

¹² The variables included in the PCA are Uninterrupted Power Supply (UPS), Laundry Machine, Air Cooler, Fridge, TV, Smart phone, Tube well, Motorcycle,

(footnote continued)

Car, bedrooms per household member, Livestock.

¹³ Measured by the share of the population in the district that belongs to the poorest two wealth quintiles.

Percentage OOSC and No. of PPP Schools

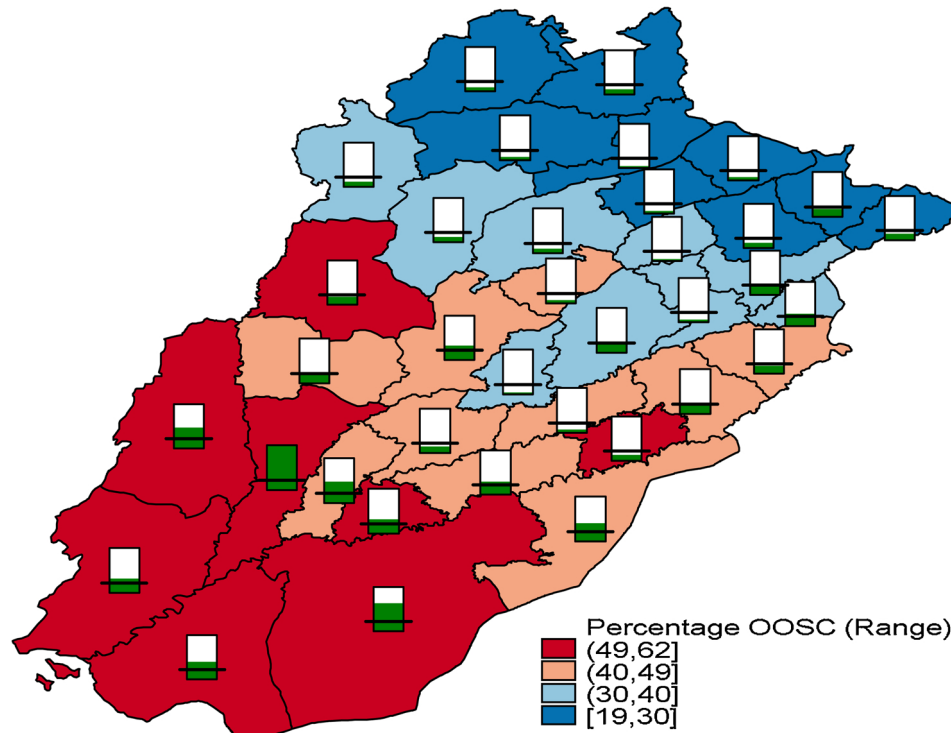


Fig. 1. Percentage of OOSC and Number of PPP Schools in District-Punjab.

Notes: Data sources: Public School Annual School Census 2017, PEF's administrative data 2017, the Private School Census Data 2016, and the MICS 2011.

Table 2a

Model estimates for type of school attended: Boys.

Variables	Model 1		Model 2	
	(2) PPP	(3) Private	(2) PPP	(3) Private
Living in southern districts	0.386*** (2.91)	-0.032 (-0.53)	0.423*** (3.84)	-0.018 (-0.33)
Age	0.009** (2.13)	-0.009 (-0.62)	0.012** (2.15)	-0.007 (-0.59)
Father Educated	0.016 (1.4)	0.016 (1)	0.020 (1.01)	0.006 (0.56)
Mother Educated	0.027*** (2.83)	0.063* (1.9)	0.021* (1.88)	0.038 (1.34)
Father works	-0.023 (-1.21)	-0.016 (-0.45)	-0.011 (-0.6)	-0.019 (-0.68)
Asset index score	-0.022*** (-4.29)	0.053*** (3.55)	-0.013* (-1.91)	0.048*** (3.13)
Parent asks child about learning			-0.075 (-1.45)	0.017 (0.5)
Student receives tutoring			0.041 (1.34)	0.034* (1.79)
Student reads books			0.098*** (3.3)	-0.026 (-0.6)
Parent talks to teachers			-0.007 (-0.28)	0.074** (2.38)
Parent expects student to attend university			0.006 (0.41)	0.087*** (2.63)
N	1,683	1,683	1,500	1,500

Notes: (a) Baseline outcome is government school attendance. (b) Only marginal effects are reported (c) Z-statistics in parenthesis ***p < 0.01, **p < 0.05, and * p < 0.1.

Table 2b

Model estimates for type of school attended: Girls.

Variables	Model 1		Model 2	
	(2) PPP	(3) Private	(2) PPP	(3) Private
Living in southern districts	0.411** (2.08)	-0.041 (-0.67)	0.420*** (2.46)	-0.022 (-0.4)
Age	0.009 (0.42)	-0.031*** (-3.36)	0.011 (0.61)	-0.027*** (-2.69)
Father Educated	-0.036 (-0.54)	0.047 (1.62)	-0.012 (-0.17)	0.024 (0.88)
Mother Educated	-0.005 (-0.09)	0.047** (2.4)	0.000 (0.01)	0.028 (1.26)
Father works	0.031* (1.75)	-0.034 (-0.67)	0.059*** (3.4)	-0.031 (-0.77)
Asset index score	-0.011 (-0.75)	0.042*** (2.81)	-0.003 (-0.24)	0.037*** (2.52)
Parent asks child about learning			-0.054* (-1.7)	0.106*** (3.06)
Student receives tutoring			-0.051 (-1.28)	0.073** (2.19)
Student reads books			0.082 (1.58)	-0.042 (-1.48)
Parent talks to teachers			-0.021 (-0.58)	0.009 (0.41)
Parent expects student to attend university			-0.026 (-0.74)	0.029 (1.06)
N	1,560	1,560	1,417	1,417

Notes: (a) Baseline outcome is government school attendance. (b) Only marginal effects are reported (c) Z-statistics in parenthesis ***p < 0.01, **p < 0.05, and * p < 0.1.

3.2. Which PPP programs are more effectively targeting districts with relatively higher shares of out-of-school children?

Table B2 presents Poisson regression estimates for individual PPP program school counts for FAS, EVS, NSP and PSSP schools using the same set of covariates. When comparing the targeting of individual PPP programs using the expanded model, it is apparent that all PPP programs are likely to have a statistically significant presence in districts with a higher share of out of school children all is equal, however the coefficient is largest for the FAS and NSP schools. The coefficient on the number of public schools in a district was found to be statistically significant for all PPP types except FAS, suggesting that FAS schools are no more or less likely to be present in districts where there is a relatively limited supply of government schools. Amongst all the PPP school types, the coefficient on rural was found to be statistically significant, and was largest for the NSP. This finding is encouraging given that the NSP is designed to target more rural areas of the province.

3.3. What are the factors associated with attending public, PPP and private schools?

Overall, public school students account for 52% of the sample used for this analysis, while PPP and private school students account for 33% and 15% respectively (Table A3). 78% of PPP students live in southern districts of the province, as compared to 26% and 30% of public and private school students respectively. The descriptive statistics suggest that private school students come from relatively more advantaged households than PPP and public school students. Of the three types of students, private school students have the highest proportion of parents that have received some education, have higher average wealth index scores, and have a higher incidence of receiving private tuition. Interestingly, children attending PPP schools, have lower average wealth index scores than children attending public schools, and have a lower proportion of fathers that have completed at least primary schooling.

In order to learn more about the relationships between child and household characteristics and school attendance, two multinomial logistic regressions were estimated separately for boys and girls with public school attendance as the base outcome category. The rationale for running separate regressions for boys and girls is to account for sex-specific differences in school type enrollment, which have been noted in previous studies (Aslam, 2009; Alcott and Rose, 2015). The first regression model excludes any potentially endogenous variables that could be affected by the type of school attended, while the second model includes these variables. Parent teacher interactions is one example of a potentially endogenous variable as it is possible that some school types may expect more frequent parent teacher interactions and therefore increased interactions may not be a reflection of more engaged parents but rather school expectations. While these concerns are acknowledged, the results from model 2 are discussed in the following section to provide insight into whether greater parent engagement and student motivation (as measured by reading habits and university aspirations) are associated with a greater likelihood of enrollment in a particular school type. Table 2a presents the model estimates for boys and Table 2b presents the model estimates for girls.

After controlling for other variables in the model, both male and female students attending PPP schools are more likely to live in southern districts relative to public school students. Interestingly, for males and females, PPP students are no more or less likely to come from higher socio-economic status households than public school students (the coefficients on the asset index score are not statistically significant). For boys, age is positively associated with the probability of being enrolled in a PPP school, and male students who reported reading books at home are 10 percentage points more likely to be enrolled in

PPP schools than those who do not read books at home. This is the only variable that potentially indicates that male PPP students are more advantaged than public school students. Girls who have fathers who work are more likely to be enrolled in PPP schools than girls whose fathers do not work. Given that there is a weak correlation between father's employment status and the student's household asset index score, this finding does not necessarily suggest that female PPP students belong to relatively higher SES households as compared to public school students. There is also no association between PPP enrollment and having parents who speak to their child's teacher, ask their child about their learning, or have university aspirations for their child.

Expectedly, there are more apparent differences between children attending private schools and public schools. Children from wealthier households are more likely to attend private school. Boys whose parents engage with their teachers are 7 percentage points more likely to be enrolled in private schools than boys whose parents do not engage with their teachers. Similarly, boys whose parents have aspirations for them to attend university are 9 percentage points more likely to enroll in private schools as compared to boys whose parents do not have university aspirations. Girls who attend private schools come from relatively advantaged backgrounds and are more likely to have parents who ask about their schooling and are more likely to receive private tuition. Unlike boys, there is no association between parental education aspirations for their girls and the probability of being enrolled in a private school.

Given that 56% of the out of school population in the province is female, and that amongst the poorest households girls are more likely to be out of school than boys, it is worth exploring whether PPP schools are more likely to target girls than public schools to address these inequities.¹⁴ As reported earlier, there is a clear sex bias in the sub-sample used for this analysis which makes it difficult to determine whether there are genuine differences in enrollment by sex in public, PPP and private schools. However, this limitation can be overcome by using the full representative sample (i.e. including students with and without SES information) and conducting a separate multinomial regression.¹⁵ The results reflect that PPP students are 35% more likely to be female than public school students. In order to put wealth scores into perspective and allow for comparisons with out of school children, another robustness check was conducted by pooling this sample with a provincially representative household survey sample from the 2014–15 Pakistan Social and Living Standards Measurement (PSLM) survey. Using comparable asset ownership indicators across the two surveys, an asset index score was estimated for this pooled sample.¹⁶ Based on this analysis, two thirds of out of school children in the province belong to the poorest two wealth quintiles.¹⁷ In comparison, only 23% of public school students, 30% of PPP school students, and 9% of private school students belong to the poorest two wealth quintiles in the province.

3.4. How do the characteristics of children enrolled in different PPP Programs compare to public school students?

In order to compare the characteristics of children attending different PPP programs and public school students, separate multinomial logistic regressions were estimated with public school attendance as the base outcome, and EVS, FAS, NSP and PSSP attendance as the

¹⁴ Statistics derived from Punjab MICS 2017–18.

¹⁵ The full representative sample consisted of 21,024 students. The regression model included the following independent variables: living in southern districts, age, and female and public school attendance was the base outcome for the dependent variable.

¹⁶ The variables used to construct this index using principle components analysis include UPS, TV, motorcycle, bicycle, car, refrigerator, and air cooler ownership.

¹⁷ The age range used for this analysis is 7 to 11 to allow for comparison with the SDI survey sample used in this study.

comparative categories for both boys and girls (Table C1).¹⁸ Given sample size limitations, only model 1 without the endogenous variables is estimated for these sub-groups.

For boys enrolled in EVS schools, none of the coefficient estimates were found to be statistically significant however, girls with less educated fathers and fathers that are working for pay were more likely to be enrolled in EVS schools.

There are more apparent differences in the backgrounds of FAS students relative to public school students particularly for boys. Boys living in southern districts are more likely to be enrolled in FAS schools than those not living in the south of the province. Interestingly, while boys with mothers who have received some education are more likely to be enrolled in FAS schools, boys from more affluent households are less likely to be enrolled in FAS schools. For girls, no covariates were found to be statistically significantly associated with FAS enrollment.

Of the four categories of PPP students, NSP and PSSP students appear to be the most disadvantaged. Both male and female NSP students are more likely to live in southern districts. Boys with educated parents are less likely to be enrolled in NSP schools however, the magnitude of this effect is relative small. Girls enrolled in NSP schools are also less likely to have educated mothers, and less likely to belong to more affluent households.

Like FAS and NSP students, boys and girls enrolled in PSSP schools are more likely to live in southern districts. Boys enrolled in PSSP schools are less likely to have fathers working for pay, less likely to have educated mothers, and are more likely to come from poor households. For girls, only father's education was negatively associated with being enrolled in a PSSP school.

As discussed earlier, it is worth exploring whether specific PPP programs are more likely to enroll female students than public schools using the full representative sample.¹⁹ The results from this analysis reflect that EVS students are 32% more likely to be female than public school students, while none of the other PPP programs are more or less likely to have female students relative to public schools.

4. Discussion

The main purpose of this study was to better understand the geographic targeting of PPP schools and the characteristics of children who attend public, PPP, and private schools in the Punjab province. In particular, this study attempted to provide evidence to policy makers to determine whether PPP programs are targeting 'deserving' populations. Previous studies have examined the characteristics of students attending different school types however unlike this study, they have not compared the characteristics of students attending public, PPP and private schools. This study attempted to answer two main research questions: *Are PPP Programs targeting districts with relatively higher shares of out-of-school children? What are the household and parental characteristics that predict whether a child attends a public school, a PPP school or a private school in Punjab?*

The findings of the district level geographic analysis suggest that PPP interventions seem to prioritize districts with relatively higher shares of out of school children. These findings are somewhat similar to the findings from Bangladesh where ROSC project schools were found to be present in poor sub-districts (Asadullah, 2016). While individual PPP programs are not exclusively mandated to operate in these low enrollment districts, the stated focus of the School Education Department is to increase enrollment rates in these districts and PPPs are a major mechanism through which these goals are to be achieved. Of the

various PPP programs, the FAS and NSP are the most likely to be present in low enrollment districts after controlling for other factors. It is not surprising that the EVS schools are not as prevalent in low enrollment districts as the NSP, because the EVS relies on the availability of multiple private schools within a locality and the availability of private schools in low enrollment districts is relatively limited. Given that there is no association between the number of private schools in a district and the share of OOSC in a district, policy makers may need to think about how to further incentivize existing private school operators to target these districts. Due to these constraints, it seems that if the objective is to increase enrollment in low enrollment districts, the NSP and FAS seem to be more appropriate models since they have managed to target low enrollment districts possibly due to the fact that they do not rely on the existence of a large private school market unlike the EVS.

The results of the multinomial logistic regressions provide interesting insight into the backgrounds of students attending public, PPP and private schools. The findings reflect that although boys attending PPP schools were more likely to have educated mothers, children attending PPP schools do not belong to more or less wealthy households than public school students. PPP schools, particularly EVS schools were also more likely to cater to female students relative to public schools. Additionally, the results do not seem to suggest that parents with higher education aspirations for their children are opting to send their children to PPP schools. At the same time, boys who read non-school related books were more likely to be enrolled in PPP schools than boys who do not read books at home. These last two findings are particularly noteworthy because they provide mixed evidence as to whether parents are choosing to send their more motivated, academically-oriented children to PPP schools. The causal direction related to the reading habits of students is unclear, given that on average, learning outcomes in PPP schools are higher than in public schools, the higher levels of literacy amongst PPP students could explain the greater inclination towards reading relative to public school students. Alternatively, it is also possible that this finding is due to higher motivation or innate ability of PPP students relative to public students.

The trend for private school students corroborates findings from previous studies (Alcott and Rose, 2015; Asadullah and Maliki, 2018), with private school students being more likely to come from more wealthy households, and receive privation tuition. Although having university aspirations for their child was positively associated with private school enrollment for boys, this was not the case for girls. Despite coming from more affluent households, parents who send their daughters to private school do not have higher expectations of their daughters to attend university as compared to girls attending public schools.

Overall, the findings from this study suggest that there is no evidence that PPP programs are 'cream skimming' by enrolling students with similar profiles as private school students. The evidence indicates that the School Education Department is targeting somewhat similar populations through PPPs and public schooling initiatives. Based on the results of this study, it may even be possible that some PPP programs such as the NSP are doing a slightly better job than public schools in targeting disadvantaged populations. However, if the objective of PPPs is to bridge the enrollment gap in the province and enroll out of school children, then the targeting of these programs would need to be further refined to target the most vulnerable populations, particularly in the bottom two wealth quintiles where the bulk of the out of school children are concentrated. In addition to geographic and poverty based targeting, a focus on increasing female enrollment may also help reduce the number of out of school children given that more than half of the out of school population is female. EVS schools in particular are more likely to have female students than public schools, and by doing so, it is possible that these schools may be enrolling children who were previously out of school. At the very least, the findings do not support the argument that the policy of expanding access to education through PPPs is subsidizing parents who would have otherwise paid to send

¹⁸ For this analysis private school students were removed from the sample. Only marginal effects are presented.

¹⁹ The full representative sample consisted of 17,959 students. The regression model included the following independent variables: living in southern districts, age, and female.

their children to a private school in the absence of a PPP program.

There are some limitations of this study that should be acknowledged. First, given the lack of socio-economic data at the sub-district level, the results of the geographic analysis are limited to the district level which is a large geographic unit (on average there are 892 public primary schools, 321 PPP schools and 594 private schools in a district). If sub-district data were available it would allow for examining the location of PPP schools relative to public and private schools at a village level for example, which would be more appropriate to evaluate the targeting of these programs within an education marketplace.²⁰ Second, it is evident that public school students for whom socio-economic data was available had slightly higher test scores than public school students for whom data was not available, suggesting that the sample used for the regression analysis may over-represent higher performing and possibly more affluent public school students. Similarly, in the PPP sample for which SES information was collected, female students were overrepresented as compared to PPP students for whom SES information was not collected. Another limitation of this study is the lack of important demand and supply side variables which have previously been found to influence school choice in similar contexts. This includes village level information, information on the supply of schools, and multiple observations within households to provide insight into within household school choice preferences. It is possible that the inclusion of these additional covariates would reduce the effect of household level factors such as household wealth and the effect of living in southern

districts which may be correlated with neighborhood or village level factors impacting school choice. Lastly, it is important to note that students attending EVS schools who were part of this sample may not necessarily be voucher recipients and may be fee paying students attending EVS schools. This is because EVS schools do not cater exclusively to voucher students as there are many fee paying students who attend these schools. Without knowing which students are voucher recipients, it is difficult to comment with certainty whether the EVS program itself is targeting more or less disadvantaged populations.

Future research could shed more insight into the school decision making process of parents opting for PPP and public schooling, by capturing multiple observations within households and information on the availability of schooling choices for children. Such research could also attempt to measure child IQ to provide more insight into whether parents are sending more able children to particular types of schools in Punjab. To learn more about the targeting of PPP programs, additional research could also examine the impact of the introduction of a PPP school in a locality on neighboring public schools. This could shed insight into whether PPP schools are displacing students from public schools to PPP schools and ultimately help inform the strategic direction of individual PPP programs moving forward.

Declaration of Competing Interest

None.

Appendix A. Descriptive Statistics

Table A1
District Characteristics.

Variable	Mean	Standard Deviation
Number of public schools	892	337
Number of PPP schools	321	285
Number of private schools	594	224
Share of children out-of-school	0.399	0.121
Population Ages 5 to 9 (in thousands)	415	240
Share Rural	0.760	0.133
Share of households in bottom 2 wealth quintiles	0.464	0.218
N	36	

Notes: Statistics are calculated using the following data sources; MICS 2011, Annual School Census 2017, PEF Administrative Data 2017, Private School Census 2016.

Table A2
Descriptive Statistics for Children with Socio-Economic Data and without.

	Public					PPP ^a	
	No SES Data Mean (sd)	SES Data Mean (sd)	<i>t</i> -value	prob	<i>df</i>	No SES Data Mean (sd)	SES Data Mean(sd)
Share living in Urban areas	0.42 (0.49)	0.33 (0.47)	6.609	0.000	2869	0.26(0.44)	0.23(0.42)
Share living in southern districts	0.17 (0.38)	0.26 (0.44)	− 7.921	0.000	2547	0.72 (0.45)	0.76 (0.43)
Age	10.03 (1.49)	9.95 (1.49)	1.220	0.223	2701	9.91 (1.39)	9.90 1.38)
Share Female	0.48 (0.50)	0.47 (0.50)	1.691	0.091	2726	0.47 (0.50)	0.51 (0.50)
Class in which joined school	0.94 (1.25)	0.89 (1.25)	1.458	0.145	2698	1.16 (1.22)	1.25 (1.27)
Urdu Test Score	0.38 (0.16)	0.39 (0.16)	− 3.923	0.000	2668	0.44 (0.15)	0.45 (0.15)

(continued on next page)

²⁰ To provide some perspective as to what constitutes a typical educational marketplace in Punjab, Andrabi et al. (2007) reported that a typical village in their study sample had 8 schools consisting of 5 government schools and three private schools.

Table A2 (continued)

	Public						PPP ^a	
	No SES Data Mean (sd)	SES Data Mean (sd)	t-value	prob	df		No SES Data Mean (sd)	SES Data Mean(sd)
Math Test Score	0.34 (0.14)	0.36 (0.15)	− 3.308	0.001	2689		0.42 (0.14)	0.42 (0.14)
English Test Score	0.37 (0.14)	0.38 (0.15)	− 2.153	0.031	2701		0.41 (0.14)	0.40 (0.14)
Listening Test Score	0.67 (0.18)	0.67 (0.18)	0.110	0.913	2742		0.72 (0.16)	0.71 (0.17)
Overall Test Score	0.38 (0.13)	0.39 (0.13)	− 3.618	0.000	2689		0.44 (0.12)	0.44 (0.12)

	PPP ^a			Private				
	t-value	prob	df	No SES Data Mean (sd)	SES Data Mean (sd)	t-value	prob	df
Share living in Urban areas	2.961	0.003	1416	0.54 (0.50)	0.41 (0.41)	5.885	0.000	1071
Share living in southern districts	− 4.290	0.000	1991	0.24 (0.43)	0.28 (0.45)	− 2.429	0.015	899
Age	0.355	0.722	1856	9.56 (1.14)	9.49 (1.14)	1.459	0.145	926
Share Female	− 1.977	0.048	1865	0.47 (0.50)	0.45 (0.50)	− 0.154	0.877	964
Class in which joined school	− 2.574	0.010	1799	0.64 (1.10)	0.74 (1.15)	− 1.278	0.202	927
Urdu Test Score	− 2.184	0.029	1851	0.43 (0.15)	0.42 (0.16)	0.118	0.906	931
Math Test Score	− 0.138	0.890	1805	0.41 (0.15)	0.40 (0.15)	0.543	0.587	944
English Test Score	0.971	0.332	1856	0.48 (0.17)	0.46 (0.16)	2.453	0.014	991
Listening Test Score	1.831	0.067	1862	0.80 (0.17)	0.78 (0.17)	1.307	0.192	1007
Overall Test Score	− 0.525	0.600	1851	0.46 (0.13)	0.45 (0.13)	1.251	0.211	950

Notes: Standard deviations reported in parenthesis. Results of *t*-test assuming unequal variances (a) PPP denotes Public Private Partnership schools.

Table A3

Descriptive Statistics for sample of Grade 4 Students in 2018 SDI Survey.

	Public % or Mean (sd)	PPP ^a % or Mean (sd)	Private % or Mean (sd)	EVS ^b % or Mean (sd)	FAS ^c % or Mean (sd)	NSP ^d % or Mean (sd)	PSSP ^e % or Mean (sd)
Living in southern districts	26	78	30	57	82	95	64
Age	9.93 (1.51)	9.90 (1.38)	9.50 (1.16)	10.18 (1.38)	9.76 (1.33)	10.22 (1.49)	9.81 (1.44)
Share Female	47	53	46	55	50	53	44
Father Educated	68	63	85	57	63	46	59
Mother Educated	51	45	74	46	46	23	40
Father works	47	45	44	55	46	37	43
Asset index score	− 0.20 (1.52)	− 0.39 (1.54)	1.18 (1.93)	− 0.16 (1.66)	− 0.45 (1.52)	− 0.87 (1.50)	− 0.57 (1.46)
Parent asks child about learning	70	63	87	64	56	51	64
Student receives tutoring	41	22	53	31	24	14	21
Student reads books	25	33	25	26	38	22	28
Parent talks to teachers	53	50	72	52	47	45	49
Parent expects student to attend university	44	50	66	47	49	33	45
Overall Percentage	52	33	15	7	7	6	7
N*	1531	954	432	196	202	187	203

Notes: (a) PPP denotes public-private partnership schools (b) EVS denotes Education Voucher Scheme schools (c) FAS denotes Foundation Assisted Schools (d) NSP denotes New School Program schools (e) PSSP denotes Public School Support Program schools (*) Individual PPP program student numbers do not tally up to the overall PPP numbers because for some students information on PPP program type was missing.

Appendix B. Poisson Regression Estimates of Determinants of School Presence in Punjab Districts

Table B1

Poisson Regression Estimates of Determinants of School Presence by School Type.

	Public		PPP ^a		Private	
	1	2	1	2	1	2
Share of children out of school	0.296*** (5.58)	−0.835*** (7.02)	0.964*** (6.52)	2.155*** (6.50)	0.832*** (12.50)	−0.217 (1.46)
Estimated number of 5-9 year old children (in thousands)	0.001*** (39.840)	0.001*** (32.490)	0.002*** (18.660)	0.000 (0.170)	0.001*** (21.710)	0.001*** (14.290)
rural	1.484*** (24.13)	1.079*** (14.03)	1.674*** (9.98)	1.69*** (7.17)	−0.199*** (0.07)	−0.805*** (8.61)
Share of population belonging to bottom 2 wealth quintiles	–	0.757*** (9.61)	–	−1.445 (6.45)	–	0.73*** (4.59)
No. of Public Schools	–	–	–	0.000*** (12.70)	–	0.000*** (4.59)
No. of Private Schools	–	0*** (4.08)	–	0.001*** (12.70)	–	–
Constant	5.001*** (97.52)	5.322*** (79.85)	2.411*** (17.38)	1.891*** (9.48)	5.827*** (98.30)	6.284*** (84.70)
N	36	36	36	36	36	36
Pseudo R ²	0.453	0.486	0.31	0.471	0.428	0.459

Notes: (a) PPP denotes Public-Private Partnership School. Absolute values of z-statistics are reported in parentheses ***p < 0.01, ** p < 0.05, and * p < 0.1.

Table B2

Poisson Regression Estimates of Determinants of School Presence by Public-Private Partnership School Type.

	EVS ^a		FAS ^b		NSP ^c		PSSP ^d	
	1	2	1	2	1	2	1	2
Share of children out of school	6.511*** (21.45)	5.316*** (9.37)	6.117*** (31.47)	7.001*** (19.30)	6.061*** (23.37)	6.406*** (13.09)	0.964*** (6.52)	2.155*** (6.50)
Estimated number of 5-9 year old children (in thousands)	0.003*** (22.530)	0.004*** (19.690)	0.002*** (31.470)	0 (0.420)	0.002*** (21.05)	0*** (3.81)	0.002*** (18.66)***	0 (0.17)
Rural	−0.616*** (2.69)	1.1** (2.42)	0.602*** (3.07)	1.311*** (4.40)	3.36*** (11.44)	2.657*** (6.13)	1.674*** (9.98)	1.69*** (7.17)
Share of population belonging to bottom 2 wealth quintiles		1.3*** (3.09)		−0.902 (3.44)		−0.246 (0.69)		−1.445*** (6.45)
No. of Public Schools		−0.001*** (11.16)		0 (0.45)		0*** (3.81)		0.001*** (11.24)
No. of Private Schools		0 (1.21)		0.002*** (19.88)		0.002*** (15.12)		0.001*** (12.70)
Constant	−0.08 (0.41)	−1.334*** (3.48)	0.684*** (4.43)	−0.65*** (2.64)	−2.38*** (10.43)	−2.781*** (7.85)	2.411*** (17.38)	1.891*** (9.48)
N	36	36	36	36	36	36	36	36
Pseudo R ²	0.679	0.746	0.587	0.704	0.7039	0.81	0.3101	0.471

Notes: (a) EVS denotes Education Voucher Scheme schools (b) FAS denotes Foundation Assisted Schools (c) NSP denotes New School Program schools (d) PSSP denotes Public School Support Program schools. Absolute values of z-statistics are reported in parentheses ***p < 0.01, ** p < 0.05, and * p < 0.1.

Appendix C. Multinomial logistic regression model for predicting PPP Program Type Relative to Public Schooling

Table C1

Model estimates for type of PPP School Attended (Boys and Girls).

Variables	Boys				Girls			
	(2) EVS ^a	(3) FAS ^b	(4) NSP ^c	(5) PSSP ^d	(2) EVS	(3) FAS	(4) NSP	(5) PSSP
Living in southern districts	0.060 (1.37)	0.200*** (2.60)	0.033** (2.11)	0.055*** (8.15)	0.054 (1.010)	0.206* (1.780)	0.041*** (2.970)	0.032** (2.040)
Age	0.007 (1.03)	0.009 (0.22)	0.001* (1.66)	-0.002 (-0.59)	0.007 (0.700)	-0.008 (-1.15)	0.001 (1.390)	-0.003 (-0.93)
Father Educated	-0.027 (-1.18)	0.008 (0.91)	-0.002*** (-3.95)	0.007 (1.08)	-0.032** (-2.25)	0.008 (0.280)	0.001 (1.610)	-0.011** (-1.96)
Mother Educated	0.016 (1.19)	0.027** (2.55)	-0.002** (-2.47)	-0.008** (-2.36)	0.004 (0.200)	-0.007 (-0.44)	-0.006*** (-3.06)	0.000 (-0.1)
Father works	0.011* (1.74)	-0.008 (-1.45)	-0.001 (-0.99)	-0.020*** (-4.45)	0.032*** (3.920)	-0.008 (-0.43)	-0.004* (-1.79)	-0.003 (1.530)
Asset index score	0.001 (0.14)	-0.005** (-2.18)	-0.001* (-1.87)	-0.005** (-2.33)	0.01 (1.380)	-0.002 (-0.31)	-0.001** (-2.02)	-0.003 (-1.28)
N	1338	1338	1338	1338	1248	1248	1248	1248

Notes: (a) EVS denotes Education Voucher Scheme schools (b) FAS denotes Foundation Assisted Schools (c) NSP denotes New School Program schools (d) PSSP denotes Public School Support Program schools (e) Baseline outcome is government school attendance. (f) Only marginal effects are reported (g) Z-statistics in parenthesis ***p < 0.01, ** p < 0.05, and * p < 0.1.

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